The first ruler — and why we retired it

HOW THE FIRST RULER WORKED

One **relative coordinate.** The pericentral- and periportal-marker programs are each z-scored across all cells; “zonation strength” is read off the spread / anti-correlation of that coordinate — one number per group.

Markers: Paper 2’s landmark genes — later the full transcriptome-wide zonated set (1,273 pericentral / 364 periportal genes).

① Depth-sensitive

z-scoring depends on cell count and sequencing depth. F4 biopsies run shallower — so the spread shrinks for non-biological reasons, even when zonation is unchanged.

② Mechanism-blind

depletion, dimming, co-expression, turn-off and noise all shrink the spread the same way. One number cannot tell them apart.

THE TELL

Biopsy-only, the relative ruler showed an apparent detox “drift” (a relative score sliding 66 → 49). It **did not survive the count-based classification** — the same donors, counted in real molecules, show nothing structural. That single number was reading depth and composition, not biology — so we replaced it with depth-matched molecule counts.

The two ends are confounded — the stress is handling

THREE TISSUE SOURCES — only one is a needle biopsy

Disease spectrum · F0–F4

16-gauge needle core biopsy (right lobe, ~2 cm)

38 donors (F0 2 · F1 8 · F2 12 · F3 12 · F4 4)

Healthy

deceased-donor organ cube (~1 cm^3, multi-lobe)

4 donors

End-stage

whole-organ explant cubes (multi-lobe)

5 donors

Both ends are surgical organ cubes — not needle biopsies. Disease is read only from the matched biopsies (F0–F4).

HANDLING, NOT HYPOXIA

The end-stage spike is the acute **immediate-early** program (IEG) — not the sustained **hypoxia** program (HIF). Fold = end-stage ÷ biopsy:

Lineage	IEG	HIF
Hepatocyte	**18.5×**	1.8×
Endothelial (non-zonated)	**18.2×**	2.6×
Stellate	**12.0×**	2.6×
Cholangiocyte	**10.6×**	0.8×
Macrophage	**5.3×**	1.9×
Lymphocyte	**3.2×**	1.6×

IEG jumps ~18× — as much in non-zonated endothelium as in hepatocytes. HIF barely moves. **Disease ischemia would lead with HIF; instead acute handling leads** → a procurement / dissociation signature, not zonation.

Documented snRNA dissociation / handling artifact: van den Brink et al. 2017 (Nat. Methods); O'Flanagan et al. 2019 & Denisenko et al. 2020 (Genome Biol.). The immediate-early / heat-shock signature is the canonical tissue-handling tell.

Raw-count anchor classification — the exact steps

raw UMI counts → down-thin to 1,500 → anchor ON if ≥ 2 UMI → classify PC/PP/dual/null → fraction per donor → median by stage

THE ANCHORS

Pericentral: GLUL, CYP3A4

Periportal: ASS1, PCK1, HAL, ALDOB

Rule (mutually exclusive): PC = ≥1 PC-anchor & <2 PP; PP = ≥2 PP & <1 PC; dual = ≥1 PC & ≥2 PP; null = neither.

WHY THESE CHOICES

Down-thin to a common 1,500 UMIs so depth can't fake a class shift (held at 1,000 / 1,500 / 3,000; Monte-Carlo SD 0.006–0.010).

≥ 2 UMI = ambient-robust (≥ 1 lets ambient RNA call a class).

Donor is the unit — nuclei never pseudoreplicated (Squair et al. 2021).

9 / 9 raw-data sanity checks pass *(src/prep/06_sanity_raw.py)*

- ✓ raw integer UMIs (not SCT)
- ✓ panel == recorded library
- ✓ raw range, not the SCT band
- ✓ differs from the SCT matrix
- ✓ ALB dominates hepatocytes
- ✓ panel ≤ library, per cell
- ✓ 69,426 hepatocytes (matches paper)
- ✓ detection rates plausible
- ✓ no silent gene drop